EXPERIMENT-3

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1. COMMAND LINE INTERFACE:

PYTHON CODE:

```
from typing import List, Dict

# --- Quiz data and shared logic ---
questions: List[Dict] = [
    {"q": "What is 2 + 2?", "a": "4"},
    {"q": "What is the capital of France?", "a": "Paris"},
    {"q": "What color do you get when you mix red and white?", "a": "pink"},
]

def run_quiz(prompt_func, output_func):
    """Run the quiz using provided prompt and output functions.

    prompt_func(question_text) -> user answer (string)
    output_func(text) -> display or speak text
    """
    score = 0
    total = len(questions)

    output_func(f"Welcome to the Simple Quiz! {total} questions.")

    for idx, item in enumerate(questions, start=1):
        q_text = f"Q{idx}. {item['q']}"
        answer = prompt_func(q_text)
        if answer is None:
            # Treat None as skipped
            output_func("No answer received; moving on.")
            continue
        if answer.strip().lower() == item['a'].strip().lower():
            output_func("Correct!")
            score += 1
        else:
            output_func(f"Incorrect. The correct answer is: {item['a']}")

    output_func(f"Quiz complete. Your score: {score}/{total}")
    return score

# --- CLI-specific prompt / output ---

def cli_prompt(question_text: str) -> str:
    try:
        return input(question_text + "\nYour answer: ")
    except (EOFError, KeyboardInterrupt):
        print()  # newline
```

```
        return None

def cli_output(text: str):
    print(text)

if __name__ == "__main__":
    run_quiz(cli_prompt, cli_output)
```

```
PS C:\Users\Nthish\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Nthish\AppData\Local\Programs\Python\Python313\python.exe /workspace/cli_quiz.py
Welcome to the Simple Quiz! 3 questions.
Q1. What is 2 + 2?
Your answer: 4
Correct!
Q2. What is the capital of France?
Your answer: paris
Correct!
Q3. What color do you get when you mix red and white?
Your answer: pink
Correct!
Quiz complete. Your score: 3/3
PS C:\Users\Nthish\AppData\Local\Programs\Microsoft VS Code>
```

2. GRAPHICAL USER INTERFACE:

PYTHON CODE:

```
import tkinter as tk
from tkinter import messagebox, ttk
from typing import List, Dict

# --- Quiz data and shared logic (same questions as other interfaces) ---
questions: List[Dict] = [
    {
        "q": "What is 2 + 2?",
        "a": "4",
        "choices": ["3", "4", "5", "22"],
    },
    {
        "q": "What is the capital of France?",
        "a": "Paris",
        "choices": ["Berlin", "Madrid", "Paris", "Rome"],
    },
    {
        "q": "What color do you get when you mix red and white?",
        "a": "pink",
        "choices": ["Purple", "Pink", "Orange", "Brown"],
    },
]

def run_quiz(prompt_func, output_func):
    """Shared quiz runner used by other interfaces.

    It expects a `prompt_func` that returns the user's answer string
    and an `output_func` to display feedback.
```

```
    """
    score = 0
    total = len(questions)
    output_func(f"Welcome to the Simple Quiz! {total} questions.")

    for idx, item in enumerate(questions, start=1):
        q_text = f"Q{idx}. {item['q']}"
        answer = prompt_func(q_text)
        if answer is None:
            output_func("No answer received; moving on.")
            continue
        if answer.strip().lower() == item['a'].strip().lower():
            output_func("Correct!")
            score += 1
        else:
            output_func(f"Incorrect. The correct answer is: {item['a']}")

    output_func(f"Quiz complete. Your score: {score}/{total}")
    return score

# --- GUI-specific code ---
class QuizGUI:
    def __init__(self, master):
        self.master = master
        master.title("Simple Quiz")
        master.geometry("620x420")
        master.configure(bg="#f4f7fb")
        master.resizable(False, False)

        # Styling
        self.style = ttk.Style(master)
        try:
            self.style.theme_use('clam')
        except Exception:
            pass
        self.style.configure('Card.TFrame', background='white')
        self.style.configure('Title.TLabel', font=(None, 20, 'bold'), background='white')
        self.style.configure('Question.TLabel', font=(None, 13), background='white')
        self.style.configure('Choice.TButton', font=(None, 12), padding=10)
        self.style.map('Choice.TButton', background=[('active', '#dfefff')])

        self.index = 0
        self.score = 0

        # Outer padding frame
        container = ttk.Frame(master, padding=20, style='Card.TFrame')
        container.place(relx=0.5, rely=0.5, anchor=tk.CENTER, width=560, height=360)

        # Title
        self.title_label = ttk.Label(container, text='Simple Quiz', style='Title.TLabel')
        self.title_label.pack(pady=(6, 10))
```

```
        # Card for question and choices
        card = ttk.Frame(container, padding=(12, 10), style='Card.TFrame')
        card.pack(fill=tk.BOTH, expand=True)

        # Question
        self.question_var = tk.StringVar()
        self.question_label = ttk.Label(card, textvariable=self.question_var, style='Question.TLabel',
wraplength=500)
        self.question_label.pack(pady=(4, 12))

        # Choices grid - larger button appearance
        self.choices_frame = ttk.Frame(card, style='Card.TFrame')
        self.choices_frame.pack(fill=tk.BOTH, expand=True)

        self.choice_buttons = []
        for r in range(2):
            for c in range(2):
                i = r * 2 + c
                btn = ttk.Button(self.choices_frame, text=f'Choice {i+1}', style='Choice.TButton',
command=lambda i=i: self.on_choice(i))
                btn.grid(row=r, column=c, padx=8, pady=8, sticky=tk.NSEW)
                self.choice_buttons.append(btn)
        for i in range(2):
            self.choices_frame.grid_columnconfigure(i, weight=1)

        # Feedback and progress
        bottom = ttk.Frame(container, padding=(6, 6), style='Card.TFrame')
        bottom.pack(fill=tk.X)

        self.feedback_var = tk.StringVar()
        self.feedback_label = ttk.Label(bottom, textvariable=self.feedback_var, foreground='#2b7a78',
background='white')
        self.feedback_label.pack(side=tk.LEFT)

        self.progress = ttk.Progressbar(bottom, maximum=len(questions), length=220)
        self.progress.pack(side=tk.RIGHT)

        # Controls
        controls = ttk.Frame(container, style='Card.TFrame')
        controls.pack(pady=(8, 0))

        self.next_button = ttk.Button(controls, text='Next', command=self.next_question,
state=tk.DISABLED)
        self.next_button.grid(row=0, column=0, padx=6)

        self.restart_button = ttk.Button(controls, text='Restart', command=self.restart)
        self.restart_button.grid(row=0, column=1, padx=6)

        # Keyboard bindings for choices 1-4 and Enter for Next
        master.bind('1', lambda e: self._kbd_choice(0))
        master.bind('2', lambda e: self._kbd_choice(1))
        master.bind('3', lambda e: self._kbd_choice(2))
```

```
    master.bind('4', lambda e: self._kbd_choice(3))
    master.bind('<Return>', lambda e: self._kbd_next())

    self.show_question()

def _kbd_choice(self, idx):
    if idx < len(self.choice_buttons) and self.choice_buttons[idx]['state'] == tk.NORMAL:
        self.on_choice(idx)

def _kbd_next(self):
    if self.next_button['state'] == tk.NORMAL:
        self.next_question()

def show_question(self):
    if self.index >= len(questions):
        self.finish_quiz()
        return

    q = questions[self.index]
    self.question_var.set(f"Q{self.index+1}. {q['q']}")

    choices = q.get('choices') or [q['a']]
    while len(choices) < 4:
        choices.append(choices[-1])

    for btn, text in zip(self.choice_buttons, choices[:4]):
        btn.config(text=text, state=tk.NORMAL)

    self.feedback_var.set(f"Question {self.index+1} of {len(questions)}")
    self.progress['value'] = self.index
    self.next_button.config(state=tk.DISABLED)

def on_choice(self, choice_index: int):
    q = questions[self.index]
    selected_text = self.choice_buttons[choice_index].cget('text')
    correct = q['a']

    for btn in self.choice_buttons:
        btn.config(state=tk.DISABLED)

    if selected_text.strip().lower() == correct.strip().lower():
        self.score += 1
        self.feedback_var.set('Correct ✅')
    else:
        self.feedback_var.set(f"Incorrect ❌ (Correct: {correct})")

    self.next_button.config(state=tk.NORMAL)

def next_question(self):
    self.index += 1
    self.show_question()
```

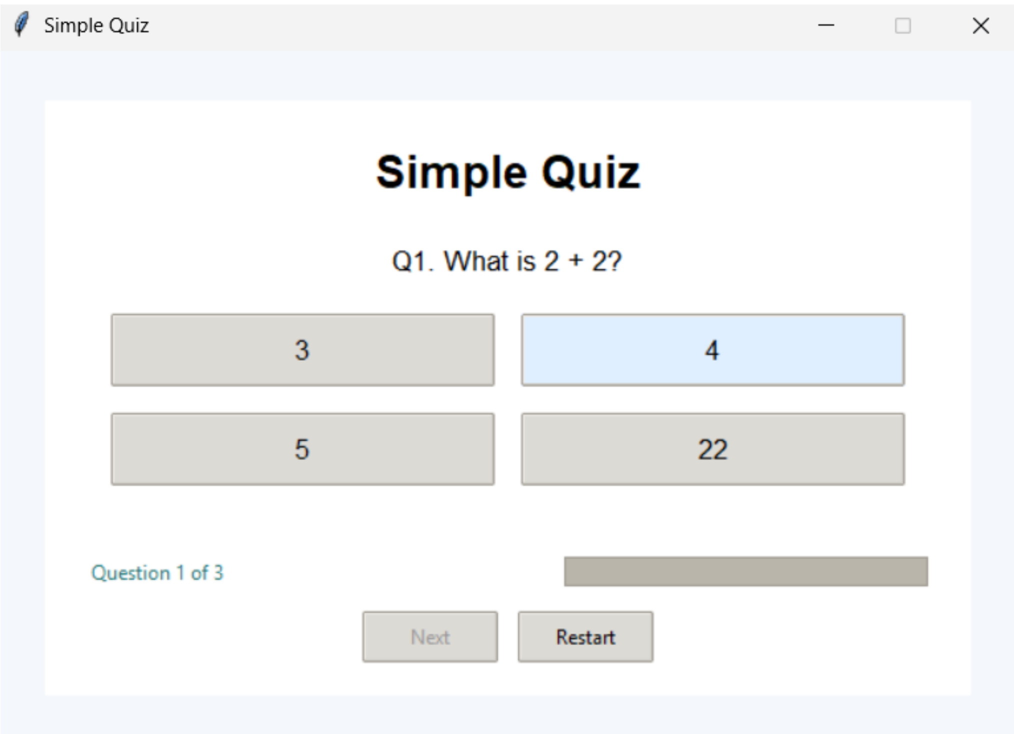
```
    def finish_quiz(self):
        self.progress['value'] = len(questions)
        # Summary window
        summary = tk.Toplevel(self.master)
        summary.title('Results')
        summary.geometry('360x180')
        summary.transient(self.master)
        summary.grab_set()

        ttk.Label(summary, text='Quiz Complete', font=(None, 16, 'bold')).pack(pady=(12, 6))
        ttk.Label(summary, text=f'Your score: {self.score}/{len(questions)}', font=(None,
13)).pack(pady=(0, 10))

        btn_frame = ttk.Frame(summary)
        btn_frame.pack(pady=8)
        ttk.Button(btn_frame, text='Restart', command=lambda: (summary.destroy(),
self.restart())).grid(row=0, column=0, padx=8)
        ttk.Button(btn_frame, text='Close', command=self.master.quit).grid(row=0, column=1, padx=8)

    def restart(self):
        self.index = 0
        self.score = 0
        self.show_question()

if __name__ == "__main__":
    root = tk.Tk()
    root.geometry("480x320")
    app = QuizGUI(root)
    root.mainloop()
```



3. VOICE USER INTERFACE:

PYTHON CODE:

```
import sys
import threading
import time
from queue import Queue

# Optional voice libs: do not raise if missing — fall back to typed I/O
try:
    import speech_recognition as sr
    _HAS_STT = True
except Exception:
    sr = None
    _HAS_STT = False

try:
    import pyttsx3
    _HAS_TTS = True
except Exception:
    pyttsx3 = None
    _HAS_TTS = False

def speak(text):
    """Speak using pyttsx3 if available, otherwise print."""
    if _HAS_TTS and pyttsx3 is not None:
        try:
            engine = pyttsx3.init()
            engine.say(text)
            engine.runAndWait()
            return
        except Exception:
            pass
    print(text)

def listen_once(recognizer, mic, seconds=3):
    """Try to listen from mic and return recognized text; fallback to typed input."""
    if _HAS_STT and sr is not None and recognizer is not None and mic is not None:
        print(f"Speak now (listening for up to {seconds} seconds)...")
        try:
            audio = recognizer.listen(mic, phrase_time_limit=seconds)
        except Exception:
            print('No speech detected (timeout or mic error).')
            return None
        try:
            return recognizer.recognize_google(audio)
        except sr.UnknownValueError:
            print('Could not understand audio.')
        except Exception as e:
            print('Recognition error:', e)
```

```
        return None
    # Fallback: typed input
    try:
        typed = input(f"(type) Your answer (or leave blank): ")
        return typed if typed.strip() else None
    except (EOFError, KeyboardInterrupt):
        print()
        return None

def interactive_loop():
    """Interactive terminal loop — uses voice if available, otherwise typed input."""
    recognizer = None
    microphone = None
    if _HAS_STT and sr is not None:
        try:
            recognizer = sr.Recognizer()
            microphone = sr.Microphone()
        except Exception:
            recognizer = None
            microphone = None

    history = []
    timer = 3

    print('\nTerminal VUI demo — voice optional')
    print("Commands: Enter=SpeakNow, 't'=set timer, 'h'=history, 's'=speak text, 'q'=quit,
'quiz'=run quiz")

    while True:
        try:
            cmd = input('\n> ').strip()
        except (KeyboardInterrupt, EOFError):
            print('\nExiting.')
            break

        if cmd == '':
            # Speak now
            if recognizer and microphone:
                with microphone as source:
                    try:
                        recognizer.adjust_for_ambient_noise(source, duration=0.4)
                    except Exception:
                        pass
                    text = listen_once(recognizer, source, seconds=timer)
            else:
                text = listen_once(None, None, seconds=timer)

            if text:
                print('You said:', text)
                history.append(text)
                handle_spoken(text)
```

```
        elif cmd == 't':
            val = input('Set timer seconds (e.g. 3): ').strip()
            try:
                v = int(val)
                if v <= 0:
                    raise ValueError
                timer = v
                print('Timer set to', timer, 'seconds')
            except Exception:
                print('Invalid number.')

        elif cmd == 'h':
            print('\nTranscript history:')
            for i, h in enumerate(history[-20:], start=1):
                print(f' {i}.', h)

        elif cmd == 's':
            text = input('Text to speak: ').strip()
            if text:
                speak(text)

        elif cmd == 'quiz':
            vui = VUI()
            def vui_prompt(q_text: str) -> str:
                return vui.listen(q_text)
            def vui_output(text: str):
                vui.speak(text)
                time.sleep(0.3)
            run_quiz(vui_prompt, vui_output)

        elif cmd == 'q':
            print('Quitting.')
            break

        else:
            print("Unknown command. Press Enter to 'Speak Now', or 'q' to quit.")

if __name__ == '__main__':
    interactive_loop()

from typing import List, Dict
import time

# Try to import optional voice libraries
try:
    import pyttsx3
    _HAS_TTS = True
except Exception:
    pyttsx3 = None
    _HAS_TTS = False

try:
```

```
    import speech_recognition as sr
    _HAS_STT = True
except Exception:
    sr = None
    _HAS_STT = False

# --- Quiz data and shared logic ---
questions: List[Dict] = [
    {"q": "What is 2 + 2?", "a": "4"},
    {"q": "What is the capital of France?", "a": "Paris"},
    {"q": "What color do you get when you mix red and white?", "a": "pink"},
]

def run_quiz(prompt_func, output_func):
    score = 0
    total = len(questions)
    output_func(f"Welcome to the Simple Quiz! {total} questions.")

    for idx, item in enumerate(questions, start=1):
        q_text = f"Q{idx}. {item['q']}"
        answer = prompt_func(q_text)
        if answer is None:
            output_func("No answer received; moving on.")
            continue
        if answer.strip().lower() == item['a'].strip().lower():
            output_func("Correct!")
            score += 1
        else:
            output_func(f"Incorrect. The correct answer is: {item['a']}")

    output_func(f"Quiz complete. Your score: {score}/{total}")
    return score

# --- VUI helpers ---
class VUI:
    def __init__(self):
        self.tts_engine = None
        if _HAS_TTS:
            try:
                self.tts_engine = pyttsx3.init()
            except Exception:
                self.tts_engine = None

        if _HAS_STT:
            try:
                self.recognizer = sr.Recognizer()
                self.microphone = sr.Microphone()
            except Exception:
                self.recognizer = None
                self.microphone = None
        else:
            self.recognizer = None
```

```python
        self.microphone = None

    def speak(self, text: str):
        """Speak the text if TTS is available, otherwise print it."""
        if self.tts_engine:
            try:
                self.tts_engine.say(text)
                self.tts_engine.runAndWait()
            except Exception:
                print("[TTS failed]", text)
        else:
            print(text)

    def listen(self, prompt: str, timeout: int = 5) -> str:
        """Try to listen from the microphone and return recognized text.

        If speech libraries are not available or recognition fails, fall back to typed input.
        """
        # Ask the user first (via voice or print)
        self.speak(prompt)
        # If speech recognition is configured, attempt to use it
        if self.recognizer and self.microphone:
            try:
                with self.microphone as source:
                    self.recognizer.adjust_for_ambient_noise(source, duration=0.5)
                    self.speak("Please speak your answer now.")
                    audio = self.recognizer.listen(source, timeout=timeout)
                # Use recognizer's default (Google Web Speech) recognizer if available
                try:
                    text = self.recognizer.recognize_google(audio)
                    return text
                except Exception:
                    self.speak("Sorry, I didn't catch that.")
                    return None
            except Exception:
                # Any microphone/recognition error -> fallback
                self.speak("Voice input is not available; please type your answer.")

        # Fallback: typed input
        try:
            # Print prompt and accept typed answer
            typed = input(prompt + "\nYour answer (type): ")
            return typed
        except (EOFError, KeyboardInterrupt):
            print()
            return None

if __name__ == "__main__":
    vui = VUI()

    def vui_prompt(q_text: str) -> str:
        return vui.listen(q_text)
```

```
def vui_output(text: str):
    vui.speak(text)
    # small pause so speech is easier to follow
    time.sleep(0.3)

# Explain capabilities to the user
if vui.tts_engine is None and (vui.recognizer is None or vui.microphone is None):
    print("Voice libraries not available. Running in text fallback mode.")
    print("To add voice support: pip install pyttsx3 SpeechRecognition pyaudio")

run_quiz(vui_prompt, vui_output)
```

```
Starting quiz: 3 questions. Press Enter to Speak Now for each question.
Q1. What is 2 + 2?
Press Enter to Speak Now...
(you press Enter, then speak "4" or type "4")
Recognized: 4
Correct!

Q2. What is the capital of France?
Press Enter to Speak Now...
(you press Enter, then speak "Paris" or type "Paris")
Recognized: Paris
Correct!

Q3. What color do you get when you mix red and white?
Press Enter to Speak Now...
(you press Enter, then speak "pink" or type "pink")
Recognized: pink
Correct!

Quiz complete. Your score: 3/3
```